



Unit 3 · Outcome 2 · Sustainable Development

Did it actually work? Evaluating environmental management.

KK08

VCE Environmental Science · Units 3 & 4 · Exam study summary

Digital Vector EnviroSci · single-topic mastery summary with worked good-vs-bad answers, exam traps, command words and practice.



“Evaluate” is a verdict, not a description

01 The command word “Evaluate” is a verdict , not a description This dot point is built on the single most demanding command word in the course. Most students lose marks here not because they don’t know the case study, but because they describe when they were asked to judge . To evaluate is to make a reasoned judgement of effectiveness or worth . Four things must be present, or it isn’t an evaluation: Judgement A clear verdict — how effective? “Highly effective”, “partially effective”, “largely ineffective”. Not “it had some impacts”. Criteria The yardst

Your criteria are the six sustainability principles

02 The yardstick Your criteria are the six sustainability principles You are not asked to invent criteria. The dot point hands them to you: judge the strategies against the sustainability principles . Turn each principle into a question you ask of every strategy. Ecological integrity & biodiversity Did it keep the ecosystem functioning and its species diverse and viable? Q: are the red gums, fish and waterbirds actually recovering? Resource-use efficiency Did it get more outcome from less resource, with less waste? Q: more “environment per litre”, or jus

No evidence, no evaluation

03 Measuring it No evidence, no evaluation “Effective” is a claim. A claim needs proof. Effectiveness is measured with indicators — things you can count or measure before and after — checked against the aim and any targets that were set. At Boorla, the monitoring program tracks indicators across all four Earth systems and the human community. Saying “the wetland got better” earns little; quoting an indicator earns the mark. Biosphere River red gum canopy health (% healthy crowns); waterbird breeding events per year; native fish (Murray cod) abundance vs

Read the scorecard like an examiner

04 Interactive · 3D Read the scorecard like an examiner Scroll back to the scorecard at the top and switch strategies on and off. Three patterns will show up every time — and each one is a sentence you can write in the exam. Strong but spiky Environmental flows score very high on ecology and intergenerational equity, but dip on intragenerational equity (irrigators lose water now). Effective, with a fairness cost. Quiet all-rounder Water-use efficiency lifts several spokes a little — it frees water (ecology) without taking it from anyone (equity). Less dr

Effectiveness is a balance, not a tick

05 The hard part Effectiveness is a balance , not a tick No strategy aces every principle. The skill in this dot point is weighing — saying which strengths and which limitations matter most, and committing to an overall verdict you can defend. Time Short-term cost, long-term gain. Salinity works and revegetation look ineffective for years, then pay off — strong on precaution and intergenerational equity , weak on immediate results. Principle vs principle Buy-backs boost the environment (ecology) but reduce farm output (intragenerational equity). You can’

Recovery takes time — drag the years

◇ ACTIVE RECALL

06 Interactive · 3D Recovery takes time — drag the years This is effectiveness made measurable. Slide through ten years and watch the wetland respond — then flip to “do nothing” to see what the strategies are actually being judged against . 3D scene Boorla wetland health · 2024 → 2034 Wetland health index — — Drag to orbit · use the slider and the scenario buttons below With management package Do nothing Flows only (no efficiency) Year 2024 Notice the lag : even the full package shows little for the first couple of years, then climbs. An evaluation writt

Same strategy, two answers — 4/4 vs 1/4

07 Worked example Same strategy, two answers — 4/4 vs 1/4 Question: “Evaluate the effectiveness of the environmental-flows strategy against the sustainability principles. (4 marks)” Watch what turns a description into an evaluation. ✗ Weak response · 1/4 Environmental flows released water from the dam into the wetland. The red gums started growing again and birds came back. This was good for the environment. Some farmers didn’t like it because they got less water. Overall environmental flows helped the wetland. Why 1/4: all description , no judgement aga



The P-E-L answer structure — make a Point, support with Evidence, then Link to the question.

Seven traps that quietly cost marks

⚠ EXAM TRAPS

08 Mark-losers Seven traps that quietly cost marks 1 Describing instead of judging You retell what the strategy did and stop. Fix: open with a verdict word — “effective”, “partially effective” — then prove it. 2 No criteria named You judge effectiveness but never against the sustainability principles . Fix: name the principle you’re testing in the sentence. 3 Verdict with no evidence “It was very effective” — says who? Fix: attach an indicator or number (canopy %, GL delivered, salinity EC). 4 One-sided Only strengths, or only weaknesses. Fix: evaluation

Command words, decoded

◆ COMMAND WORDS

09 Decoder Command words, decoded The verb tells you the job. Read it first, every time — it decides how high you have to climb. describe State features/characteristics. No judgement. The floor. explain Give reasons — how or why something happens. analyse Break into parts; show how they relate and interact. compare Identify similarities and differences between two things. evaluate Judge effectiveness/worth against criteria, weigh both sides, conclude. The summit of this dot point. to what extent A scaled evaluation — “fully / largely / partially / not”.

Build a full-mark evaluation, one piece at a time

10 Writing trainer Build a full-mark evaluation, one piece at a time This is the P·E·L trainer in its evaluation form. A top evaluation sentence has five moves. Tap each move to assemble a model answer and see how the parts lock together — then write your own to the same shape. P·E·L · evaluate Strategy under review: water buy-backs J · Judgement State how effective — a verdict word. C · Criterion Name the principle you're judging against. E · Evidence Quote an indicator / number. W · Weigh Add the other side — a limitation. C · Conclude Commit to an ove

Five exam questions · 20 marks

★ PRACTICE & SELF-MARK

11 Practice · Five exam questions · 20 marks Write your answer first, then reveal the weak and strong responses and the marking guide. Mark yourself honestly — the dock bottom-right keeps your running total. Q1 Distinguish between an effective and an efficient environmental management strategy. Use the Boorla case study to illustrate the difference. 2 marks Weak answer Strong answer Marking guide X Weak · 0–1/2 An effective strategy is a good one that works well and an efficient one is also good and saves money. Environmental flows are both. Why: circula

Indicator	2024 (before)	2029 (after)
Red gum canopy health	35%	68%
Waterbird breeding events / yr	0	3
Native fish (% of catch)	22%	41%
Carp (% of catch)	78%	59%

★ Good vs bad worked answers

Study the contrast — the same question answered two ways. The strong column shows the moves that earn the marks.



The P-E-L answer structure — make a Point, support with Evidence, then Link to the question.

✗ WEAK ANSWER

Environmental flows released water from the dam into the wetland. The red gums started growing again and birds came back. This was good for the environment. Some farmers didn't like it because they got less water. Overall environmental flows helped the wetland. Why 1/4: all description, no judgement against criteria. "Good" and "helped" aren't verdicts. No indicator/evidence, no named principles, no weighing. It states two sides but never decides between them.

✓ STRONG / MODEL ANSWER

Environmental flows were highly effective against ecological integrity but only partially effective on intragenerational equity. Timed releases that mimicked spring floods lifted river red gum canopy health from 35% to 68% healthy crowns and triggered the first waterbird breeding event in nine years — strong evidence the ecosystem is functioning again, also upholding intergenerational equity by securing the wetland's future. However, the water came at the expense of irrigators, who absorbed the cost while the wider community gained, straining intragenerational equity and user-pays. On balance the strategy is effective, because ecological recovery is measurable and durable, but its fairness

✗ WEAK ANSWER

Environmental flows released water from the dam into the wetland.

✓ STRONG / MODEL ANSWER

Effective means the strategy achieves its intended outcome — e.g. environmental flows restoring red gum health. Efficient means achieving that outcome with the least resource or cost — e.g. water-use efficiency upgrades delivering the same farm output with less water, freeing volume for the wetland without extra extraction. Why: a clean definition of each, plus a distinct Boorla example showing the contrast.

✗ WEAK ANSWER

The red gums started growing again and birds came back.

✓ STRONG / MODEL ANSWER

Water buy-backs were effective for intergenerational equity : purchasing permanent entitlements secured ~25 GL/yr of guaranteed environmental water , protecting the wetland for future generations regardless of drought. However, they were only partially effective on intragenerational equity — the lost production reduced irrigator incomes and flowed on to Connawarra's economy , so the burden fell on one group while benefits were shared by all. On balance the strategy is effective, because it delivers a durable, certain environmental gain, but its equity cost means it works best alongside efficiency measures that reduce how much water must be removed from production. Why: two named principles,

✗ WEAK ANSWER

Some farmers didn't like it because they got less water.

✓ STRONG / MODEL ANSWER

Environmental flows were largely effective in upholding ecological integrity. Red gum canopy health nearly doubled (35% → 68%) and waterbird breeding resumed (0 → 3 events/yr) , showing the ecosystem's structure and breeding function are recovering. Native fish also rose from 22% to 41% of the catch. However, recovery is incomplete — carp still dominate at 59% , so the community is healthier but not yet restored. Overall the flows are effective for ecological integrity within this timeframe, with the residual carp problem indicating flows alone are not sufficient and should be paired with carp control. Why: verdict, multiple indicators quoted directly, the carp limitation weighed, a conclusi

✗ WEAK ANSWER

An effective strategy is a good one that works well and an efficient one is also good and saves money. Environmental flows are both. Why: circular ("good... works well"), no real distinction, no case-study illustration.

✓ STRONG / MODEL ANSWER

The package upheld intragenerational equity only partially. Cultural flows and co-management with the Boorla people advanced equity by restoring access to culturally significant sites that had been lost — a group previously excluded from decisions. However, irrigators carried much of the cost through reduced allocations and buy-backs, while the ecological and recreational benefits were shared community-wide, so the burden was not distributed in proportion to the benefit. Efficiency upgrades and the recycled-water scheme softened this by reducing how much had to be taken from farms. On balance, equity improved for Traditional Owners and the wider community but at a concentrated cost to irriga